

AI-POWERED CAREER READINESS PLATFORM

SkillBridge AI

"From Skill Gaps to Career Opportunities"

Live App: <https://prakalyapbl-create.github.io/skillbridge-ai/>

Target Users: College Students & Fresh Graduates | **Languages:** Tamil, English, Hindi

01. Problem Statement

1. Skill Ambiguity

78%+ of engineering students do not know what specific technical skills they are missing for internships (e.g. Software Developer Intern).

2. Resource Disconnection

Students get trapped in unguided YouTube tutorials without structured weekly roadmaps or trusted source verification.

3. Regional Language Barrier

Students comfortable in Tamil or Hindi lack structured guidance explaining complex tech concepts in their native language.

4. Fake Keyword Stuffing

Students resort to fake online resume generators, leading to immediate rejection during technical interviews.

02. The SkillBridge AI Continuous Loop

SkillBridge AI creates a continuous 7-step career development cycle for college students:

Step	Module	Core Capability
1	STUDENT PROFILE	Collects degree, year, verified skills & resume upload without fabrication.
2	SKILL GAP ANALYZER	Compares profile vs target role. Categorizes GREEN, YELLOW, RED skills.
3	LEARN NAVIGATOR	Curated trusted resources in Tamil, English, and Hindi with verification badges.
4	WEEKLY ROADMAP	Step-by-step weekly milestone targets syncing live with readiness score.
5	PRACTICE PROJECTS	Practical portfolio project suggestions with exact resume bullet guidance.
6	RESUME STUDIO	Mode A Builder + Mode B Resume Tutor AI & ethical job description tailoring.
7	MATCH & APPLY	Devpost hackathons, Unstop challenges & official company career portal matches.

03. Full-Stack Technical Architecture

Layer	Technology	Implementation Details
Frontend & UI	Next.js 14 App Router + Tailwind CSS	React full-stack framework with Static Export capability, TypeScript & Lucide icons.
AI Logic Engine	Deterministic AI Helper Layer	Contextual reasoning engine supporting Tamil (ta), Hindi (hi), and English (en).
Python Backend	FastAPI Microservice	Standalone Python server in backend/main.py for structured resume analysis.
CI/CD & Deploy	GitHub Actions -> GitHub Pages	Automated deployment pipeline building static HTML/JS output to out/.

04. Demo Showcase & Product Impact

Student Profile: Arun Kumar

- **Degree:** B.E. Computer Science & Engineering (2nd Year)
- **Target Goal:** Software Developer Intern | **Language:** Tamil (■■■■■)
- **Verified Skills:** Python, Java, HTML, CSS, Basic React
- **Skill Gaps Identified:** SQL & Databases, Data Structures (DSA), Git, REST APIs

Result & Workflow Impact:

Arun completes his 4-week SQL roadmap, builds a Student Attendance REST API project, refines his resume bullet points via Resume Tutor AI, and matches Zoho Corporation SDE Intern requirements with an **85% readiness score**.

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